New Interactive Visual Tools and Statistical Methodology for Selecting and Evaluating Nonlinear Dimension Reduction Layouts of High-Dimensional Data

Response to examiners

I thank the examiners for their careful assessment of my thesis. What follows is a point by point response to the examiners' comments.

Examiner A: Prof Christian Hennig

The thesis treats the evaluation of two-dimensional representations of high-dimensional data by nonlinear approaches such as UMAP and t-SNE. The core is the use of hexagonal binning of the 2-d representation, which can be lifted into the higher dimensional space, used to visualise the relation between the 2-d representation and the original data using a tour, and to assess the quality of the fit by the so-called hexbin error (HBE). This can in particular be used to compare different representations. The thesis presents various illustrations and visualisations based on this approach. Chapter 2 presents the basic definitions and methods. Chapter 3 presents a software package to implement these. Chapter 4 presents another package that can generate benchmark datasets from an impressive amount of different models generating geometrically structured data. This can in particular be used to generate data with different clusters, potentially of very different kinds. Chapter 5 contains an empirical study in which test persons were asked to guess whether a low-dimensional representation is of the same dataset that they could see in a tour. In particular it was investigated in which way this depends on separation between clusters for a variety of representation models. Chapter 6 provides a Shiny App that allows a user to use the techniques proposed in Chapter 2 to assess different representations of their own data in an interactive manner. Chapter 7 concludes the thesis.

The thesis is very well written. The material presented is original, useful, timely, and of wide interest. As far as I see, it is also correct. The software seems well organised, well documented, and well thought through.

As the work is collaborative, I am not in a position to assess the candidate's individual contributions precisely, however I will assume that the candidate did much of the work and is willing to take responsibility for all of it.

Assuming this, I confirm that

- the thesis makes an original and substantial contribution to the discipline or area of professional practice;
- the candidate is able to critically reflect on, and engage with, complex ideas to create new knowledge and understanding;
- the candidate has presented a thesis which, in format and presentation, is appropriate to the standard expected of the degree.

I have a number of requests for amendments. In particular, the work touches upon some rather profound issues that the candidate currently treats rather naively, and that deserve more acknowledgment and discussion.

Therefore I recommend a pass with minor edits required. Chances are the edits can be made in 1-2 months. What I write below is quite long, but much of this is just to explain the issue. I'm asking for acknowledgment and some thoughts regarding these issues, not for any rewriting, much change of what is there, or much extra work.

Major issues:

General The HBE is used as a general metric in order to evaluate representations, and it is stated in several places that a representation with a better HBE "performs better". This is far from clear. In fact there are many statistics around that can be interpreted as evaluating the quality or lower dimensional representations. Actually many methods providing such representations are based on optimising an objective function that is meant to measure representation quality in some sense, starting from the variance criterion in PCA, various versions of stress in MDS, trustworthiness etc. Even t-SNE and UMAP can be written down as optimisation problems. Section 2.4 mentions and uses a number of other metrics, but formal definitions are not given, and there is no discussion regarding the pros and cons of these metrics. Given that this is so, why could HBE be expected to formalise the quality of a representation any better than the metrics that already exist? Actually HBE has the rather obvious disadvantage that it depends on the parameters of the grid, which is not the case for most other existing metrics. So what is its actual advantage? Before recommending a software to the user that assesses "performance" by means of HBE, this surely needs to be addressed.

I agree that many well-established metrics exist for assessing low-dimensional representations, each reflecting different aspects of structure preservation, and that NLDR methods themselves are typically defined through optimization objectives. Our intention is not to claim that HBE provides a universally better or more principled measure of embedding quality than these existing approaches.

I have revised the thesis to clarify that HBE serves a different role: it is a model-based, diagnostic measure that evaluates how well a fitted representation (via the hexagonal binning model) approximates the data in the original high-dimensional space. In contrast to metrics such as trustworthiness, stress, or RNX-based measures, which summarize specific aspects of local or global structure preservation, HBE is designed to be used in conjunction with visual tools (e.g., tour overlays) to help localize and interpret distortions.

I also acknowledge that HBE depends on grid parameters, and have revised the text to discuss this explicitly. Rather than viewing this as a drawback alone, I emphasize that examining HBE across a range of parameter values provides insight into the stability of a representation. I have softened language that previously suggested HBE measures "performance" in a general sense, and instead frame it as a complementary diagnostic tool alongside existing metrics.

General In many places the aim seems to be to find the "best" representation. But data analysts often use many visualisations of the same data because this can bring additional insight. Different kinds of visualisation can highlight different features of the data, and most insight can come from combining what can be learnt from several displays. So it should at least be acknowledged that bringing together a number of representations could be more informative for data analysis than looking for a single "best" one.

I agree that, in practice, examining multiple visualizations is often more informative than selecting a single "best" representation, as different layouts can highlight different aspects of the data. I have revised the thesis to avoid the term "best layout" and instead refer to identifying the most reasonable representation in a given context. I also emphasize that our approach is intended to support comparative and diagnostic analysis across multiple layouts, rather than to prescribe a single optimal choice.

General Cluster analysis is quite prominent in the thesis, and it is apparently implicitly always assumed that every dataset has a uniquely true clustering. There are several references to the "true clusters" without ever defining what these are supposed to be. This is very naive. There is no unique and agreed definition of what a cluster is, and several different legitimate clusterings can exist in the same dataset. See in particular

Christian Hennig (2015) What are the true clusters? Pattern Recognition Letters 64, 53-62, https://doi.org/10.1016/j.patrec.2015.04.009.

So if there is talk of "true clusters", there would need to be a definition of what this means, either (preferably) as functional of a general probability distribution, or using a data-based definition.

The implicit assumption that the true clusters are unique and clear is particularly implied in Section 4 where different "shapes" of clusters can be combined in a single dataset. I'd like to challenge the author to give a general definition of true clusters according to which the true clusters are indeed in all these situations those that are implied to be the true cluster here, but I'm afraid chances are that this would be rather hopeless. Appeal to intuition will not do, and neither will be a statement that "different clusters are generated by different distributions" because any mixture distribution can be split up in an infinite variety of ways into different distributions. So the best I can ask for is an informal but insightful discussion of the issue, including a clear acknowledgment that the thesis uses an informal and intuitive cluster concept that can in all likelihood not be made precise and may very well involve ambiguity. There is more than one legitimate clustering of a dataset, so it cannot be fully clear what the "true clusters" are that are mentioned.

The issue is particularly relevant when benchmarking clustering methods (Sec. 4.4.2). All results are relative to the assumed truth. Note that methods like kmeans can be seen as coming with their own implicit cluster concept, and they will deviate obviously and legitimately from clusters that are defined in a different way, as are the clusters here.

I agree that clustering does not admit a unique "true" solution in general.

In this paper, "true clusters" refers to the labels defined by the data-generating process, where each geometric component is generated separately and assigned a cluster identity. This is an operational definition used for simulation and benchmarking, rather than a universally valid notion of clustering.

I revised the thesis to clarify this point, explicitly acknowledge that multiple valid clusterings may exist for the same data, and note that results are conditional on this construction-based definition.

General In examples, PCA is usually run on high-dimensional data to reduce the dimension before applying the methods of interest to then only reduce the 10 (or so)-dimensional PCA representation to two dimensions. There should be a comment on this, why this is done and whether this is generally recommended. As far as I know, the literature on UMAP and other techniques treated here doesn't recommend this and doesn't suggest that this would be necessary.

In some examples, PCA is used as a preprocessing step to reduce the dimensionality of the data before applying NLDR methods. This is not required for methods such as UMAP or t-SNE, which can operate directly on the original high-dimensional data. In our case, PCA is used to reduce the data to a moderate dimension (e.g., 10 components) to facilitate visualization using tours. Tour-based visualization becomes increasingly difficult to interpret as dimensionality grows, and reducing the dimension allows clearer inspection of the fitted model and its structure. I emphasize that this step is used solely for visualization purposes and may influence the resulting embedding, particularly with respect to local structure.

Sec. 4.4.2 Much about this Section is quite naive and not very insightful. References for standard cluster analysis methods that are around for 50 years and more are given from between 2002 and 2019; original references are preferred. There are lots of hierarchical methods and it is not specified which one is used here. Also the concept of model-based clustering is very general and although I can guess what the author did, the given description is insufficient (not only Gaussian distributions can be mixed to define a clustering model). Given the variety of available covariance matrix models it seems very strange that the authors picked the one model that is almost identical to k-means when comparing with k-means. This is hardly ever of interest in practice, because otherwise one could stick to k-means in the first place, and the superior flexibility of model-based clustering would be wasted.

A number of cluster validity statistics are computed, but there isn't much discussion of their meaning and whether or why some of these might be more appropriate than others. In fact some of these are not even meant to be used on their own, see

Akhanli S.E., Hennig C. (2020). Comparing clusterings and numbers of clusters by aggregation of calibrated clustering validity indexes. STATISTICS AND COMPUTING, 30(5), 1523-1544, doi:10.1007/s11222-020-09958-2

Some guidance on good practices for benchmarking in cluster analysis:

Van Mechelen, I., Boulesteix, A.-L., Dangl, R., Dean, N., Hennig, C., Leisch, F., Steinley, D., & Warrens, M. J. (2023). A white paper on good research practices in benchmarking: The case of cluster analysis. WIREs Data Mining and Knowledge Discovery, 13(6), e1511. https://doi.org/10.1002/widm.1511

I revised Section 4.4.2 to improve clarity and rigor. Specifically, we will (i) add original references such as Stuart Lloyd (1982) and Joe H. Ward Jr. (1963), (ii) explicitly state that hierarchical clustering uses Ward's linkage, (iii) clarify that model-based clustering is implemented via a Gaussian mixture model with the "EII" covariance structure and justify this as a baseline comparable to k-means, and (iv) expand the discussion of cluster validity indices, including their limitations and appropriate use, drawing on guidance from Seda E. Akhanli and Christian Hennig (2020) and related benchmarking recommendations. I also clarified that this comparison is illustrative rather than a comprehensive benchmarking study.

Chapter 5 There are some problematic issues with this survey. As far as I understand, many high-dimensional datasets can lead to the same 2-d representation using the techniques discussed here. This means that participants can at best know that it is possible that datasets are the same (2-d representation and tour), but it is always just as well possible that datasets are different. Even with the best understanding of the methods and the best intuition this can never be excluded. Saying "same" with confidence is therefore always wrong, because one can guess "same", but confidence is never justified. Furthermore, I'd think that for the use of techniques such as UMAP, t-SNE etc. some understanding and training is required. They are not necessarily meant (or at least shouldn't be meant) to be used in a purely intuitive way without any idea of their inner workings. But as far as I understand, this kind of intuitive use is what the survey investigates. There should be some discussion of to what extent this is appropriate and informative.

I agree that a given 2-D NLDR representation is not uniquely determined by a single high-dimensional dataset, and therefore participants cannot definitively conclude that two visualisations arise from the same underlying data. Our intention, however, is not to assess whether participants can make a logically certain identification, but rather to study perceptual consistency: whether different visualisations of the same data are interpreted as similar or *different by human observers. In this context, a "same" response reflects perceived similarity rather than a claim of uniqueness.*

I also agree that NLDR methods such as t-SNE and UMAP typically require expertise for careful interpretation. However, these visualisations are often used in exploratory settings where users rely heavily on visual intuition. Our study is designed to evaluate how such intuitive interpretations align with the underlying data structure.

Minor issues:

p. 6 and elsewhere "NLDR can hallucinate wildly" - without a formal definition of what true structures are and how they should or should not be represented, it isn't clear what "hallucinate" means. Also arguably the methods just do what they are defined to do; without a human analyser misinterpreting the display there can be no "hallucination".

I agree that the term "hallucinate" was imprecise and potentially misleading. Our intention was to describe situations where NLDR embeddings distort the original geometry in ways that may suggest structure (e.g., strong separation between groups) that is not supported by the high-dimensional relationships. I have revised the thesis to replace this phrasing with a more precise description of geometric distortion and the potential for misinterpretation by the analyst.

p. 8 Scale the data - I worry that there may be robustness issues with the min/max scaling here. A single large outlier on a variable will imply that all other observations have hardly any distance between each other on this variable.

I agree that min–max scaling can be sensitive to extreme values and may compress the majority of observations when outliers are present. I have revised the thesis to acknowledge this limitation and now explicitly suggest more robust alternatives, such as quantile-based scaling or preprocessing steps to mitigate the influence of outliers. I retain min–max scaling as a simple default in well-behaved settings.

p. 9 Write the linear optimisation problem down explicitly. This is not transparent.

I have revised the thesis to explicitly state the linear optimization problem used to determine b_2*, including the objective and constraints. I also clarified how Equation is obtained by substituting the smallest feasible value of* a_1 *and solving for* b_2*.*

p.11 What does "RNX curve" mean?

I have clarified the meaning of the "RNX curve" by explicitly defining it as a co-ranking–based measure that evaluates neighborhood preservation across a range of neighborhood sizes, and explained how the ARNX summary is computed from this curve.

Fig. 2.5 y-axis label and annotation are missing.

In Fig. 2.5, the y-axis does not represent a meaningful variable; points are jittered vertically only to reduce overplotting and to better display the distribution of residuals.

p. 16, l. -6 “high correlation value” between what and what?

I have clarified that the “correlation” refers to the Spearman correlation between pairwise distances in p-D and 2-D as shown in the Shepard diagram. The text has been revised to make this explicit and to better explain how this value can be inflated in the presence of clustered structures.

Fig. 2.8 bottom The difference between the different metrics in assessing the layouts is staggering. This deserves a discussion.

I agree that the differences between the metrics are substantial and deserve clearer interpretation. I have revised the caption and accompanying text to emphasize that these differences reflect a trade-off between local and global structure preservation. I also clarify that our assessment is based on stability of HBE across a range of binwidths, rather than a single optimal value, and avoid presenting any layout as universally “best.”

Fig. 2.12 Layout f is called “best choice” but it doesn’t have the best HBE values where the HBE values are best (at low average bin count). What is the rationale here?

I agree that layout a performs well at very small binwidths, while layout f performs better at moderate binwidths. Our conclusion is based on performance across a range of binwidths, rather than at extreme values where HBE can be less informative.

p. 59 More published data generators for clustering are:

Maitra, R., & Melnykov, V. (2010). Simulating Data to Study Performance of Finite Mixture Modeling and Clustering Algorithms. Journal of Computational and Graphical Statistics, 19(2), 354–376. https://doi.org/10.1198/jcgs.2009.08054

C. Shand, R. Allmendinger, J. Handl, A. Webb and J. Keane, ”HAWKS: Evolving Challenging Benchmark Sets for Cluster Analysis,” in IEEE Transactions on Evolutionary Computation, 26 (6), 1206-1220, 2022, doi: 10.1109/TEVC.2021.3137369

I have added these.

p. 62 I haven’t exactly understood how the “background noise” works. Better explain such things formally and unambiguously.

I have revised the description of background noise to provide a clearer and more formal explanation, including how it is generated and incorporated into the dataset.

p. 64 “they do not produce a formal multicluster dataset” - what is that?

By “formal multicluster dataset”, I mean a dataset composed of multiple distinct and separable groups (clusters), where each cluster represents an independent structure, typically with clear boundaries or low-density separation between them.

In contrast, the branching structures generated here form a single connected object with multiple arms, so they do not consist of independent clusters despite having visually distinct components.

p. 68, l.-9 There's a bracket too many.

I removed the additional bracket.

p. 70, l.-9 Explain "Sierpinski-like".

By "Sierpinski-like", I refer to a structure created by repeatedly taking midpoints between points and the corners of a shape. This process produces a pattern with a repeating structure and visible gaps, similar to the well-known Sierpiński triangle and its higher-dimensional versions.

I use the term "-like" to indicate that the dataset only approximates this pattern using a finite number of points, rather than forming an exact mathematical fractal.

Chapter 5 I think results for PCA should be included. They would certainly be of interest, and not doing them because of some speculation what the result may be isn't convincing.

I agree that PCA can serve as a useful baseline. However, our study focuses specifically on nonlinear dimensionality reduction methods and how their transformations affect perceptual interpretation. Since PCA is a linear method that preserves global structure without introducing nonlinear distortions, it is generally consistent with the linear projections shown in the tour and therefore does not provide additional insight into the types of distortions I aim to study. For this reason, I have not included PCA in the main analysis, but I acknowledge its role as a baseline.

p. 91 BW ratio treats the clusters as spherical, not just elliptical.

I agree that the between–within (BW) ratio is most naturally suited to spherical cluster assumptions, rather than general elliptical shapes. I revised the text as: Measuring distance between clusters is classically done using the between–within (BW) ratio, which captures global separability under assumptions of roughly spherical cluster structure.

p. 92 Explicitly introduce the terminology "SAME" and "DIFFERENT" trials. Don't leave guesswork to the reader, not even if names are expressive.

I added the text as: The experiment consists of two types of trials: SAME trials and DIFFERENT trials. In SAME trials, two visualizations (one NLDR layout and the tour) are generated from the same underlying dataset, but with controlled variations in cluster separation. In DIFFERENT trials, the two visualizations are generated from different datasets.

p. 92 What qualifies data to be "attention check data"? Gaussian variance is specified as a single number, but in order to define the Gaussian's properly you need to state their covariance matrices.

I have clarified the definition of the attention-check datasets in the thesis. Specifically, I now explain that these datasets were designed to have simple and clearly separable cluster structures, making them easy to recognise and thus suitable for verifying participant attention and task understanding.

I have also revised the description of the Gaussian clusters to explicitly state the covariance structure. Each cluster is now defined using an isotropic covariance matrix of the form ($\sigma^2 I$), where ($\sigma^2 = 0.1$), ensuring equal variance across all dimensions and no correlation between variables.

p. 95 "The BW-ratio. . ." is not a sentence.

I corrected the sentence.

Sec. 5.3.7 I think that once the same dataset (or generation rule) has been used more than once, the model needs a dataset effect, probably another random effect. Results of different test persons on the same data are dependent for sure.

I acknowledge that responses corresponding to the same dataset may introduce dependence. In our study, the number of repeated observations per dataset was limited, and our primary focus was on differences across methods rather than dataset-specific effects. While I did not include a dataset-level random effect in the current model, I recognise this as a potential extension and have noted it as a limitation in the revised thesis.

Fig. 5.4 and elsewhere Estimated probability results look generally quite weak given that there are only two response options. Comment!

While the task involves binary responses ("yes"/"no"), the estimated probabilities reflect substantial uncertainty in participants' perceptual judgments rather than weak model performance. The stimuli were intentionally designed to include challenging cases (e.g., similar cluster structures, small separations, or nonlinear distortions), where distinguishing between visualisations is difficult. As a result, responses are often close to chance level, leading to moderate estimated probabilities.

Fig. B.10b How is what is shown here connected to normal distribution of the random effects? The assumption is not about the plain "correct proportion".

I agree that the normality assumption in the model applies to the random effects, not to the observed proportions of correct responses. The figure was intended as a descriptive summary of variability across subjects rather than a formal assessment of model assumptions. I have revised the text to remove the implication that this plot reflects the normality of the random effects and clarified its descriptive role.

B.8.1 "Subjects who exceeded the average time of 5-10 minutes were excluded, as determined from the pilot study." Why? I don't see any reason for this.

I agree that excluding participants based solely on longer completion times requires justification. Our intention was to remove cases where unusually long durations likely reflected interruptions or disengagement rather than task-related effort. I have clarified this rationale in the thesis.

"Finally, the collected data set is further refined by filtering out all the responses which showed the same data structures in 2-D NLDR plot and tour." I suspect this sentence doesn't say what you want to say here. In any case it is very strange.

I agree the original sentence was unclear. Our intention was to refer to responses from trials where both visualisations showed the same underlying data, rather than filtering based on participants' answers. I have revised the sentence to make this explicit.

Examiner B: Prof Anne M. Ruiz

No comments.